

Advance Analytics Using Statistics

20-08-2022

1. _____ summarizes past raw data putting it into a form usable by humans.

Answers

1. Descriptive analytics
 2. Predictive analytics
 3. Diagnostic Analytics
 4. Prescriptive Analytics
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2. _____ are used for root-cause analysis to understand why something happened.

Answers

1. Descriptive analytics
 2. Predictive analytics
 3. Diagnostic Analytics
 4. Prescriptive Analytics
-

3. Uniform distribution is _____?

Answers

1. Two discrete outcomes
 2. Is a distribution of the number of trials needed to get the first success in the repeated Bernoulli trials
 3. Number of successes per unit of time over the course of many units
 4. The probabilities of each outcome are equally distributed across the sample space
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4. What kind of distribution is flipping a coin?

Answers

1. Geometric distribution
 2. Binomial distribution
 3. Poisson distribution
 4. Uniform distribution
-

5. _____ number of successes per unit of time over the course of many units

Answers

1. Geometric distribution
 2. Binomial distribution
 3. Poisson distribution
 4. Uniform distribution
-

6. The central limit theorem (CLT) states that _____

Answers

1. The distribution of sample means approximates a normal distribution as the sample size gets larger, regardless of the population's distribution
2. The distribution of sample means approximates a normal distribution as the sample size gets smaller, regardless of the population's distribution
3. The distribution of sample means approximates a geometric distribution as the sample size gets larger, regardless of the population's distribution

4. The distribution of sample means approximates a binomial distribution as the sample size gets smaller, regardless of the population's distribution

7. Which of the following is true?

Answers

1. The standard normal distribution, also called the z-distribution, is a special normal distribution where the mean is 1 and the standard deviation is 0.

2. The standard normal distribution, also called the z-distribution, is a special normal distribution where the mean is 1 and the standard deviation is 1.

3. The standard normal distribution, also called the z-distribution, is a special normal distribution where the mean is 0 and the standard deviation is 0.

4. The standard normal distribution, also called the z-distribution, is a special normal distribution where the mean is 0 and the standard deviation is 1.

8. A cupboard A has 4 red carpets and 4 blue carpets and a cupboard B has 3 red carpets and 5 blue carpets. A carpet is selected from a cupboard and the carpet is chosen from the selected cupboard such that each carpet in the cupboard is equally likely to be chosen. Cupboards A and B can be selected in 15 and 35 ways respectively. Given that a carpet selected in the above process is a blue carpet, find the probability that it came from the cupboard B.

Answers

1. 2/5

2. 15/19

3. 31/73

4. 4/9

9. Random sampling is ____.

Answers

1. Every member of a population has an equal chance of being selected

2. Sampling ensures that groups within a population are adequately represented

3. The size of each sample is based on the size of the group relative to the population

4. samples are usually much bigger than populations, there's a no chance that entire demographics might be missed

10. It describes the important characteristics/ properties of the data using the measures the central tendency like mean/ median/mode and the measures of dispersion like range, standard deviation, variance etc.

Answers

1. Descriptive Statistics

2. Inferential Statistics

3. Both A and B

4. None of the above

11. What is correlation?

Answers

1. Correlation is mean squared difference between each data point and the centre of the distribution measured by the mean
 2. Correlation is a data point that differs significantly from other observations
 3. Correlation is a statistical measure that expresses the extent to which two variables are linearly related
 4. Correlation is statistical tool that is used to determine the relationship between the movements of two random variables.
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12. What is covariance in statistics?

Answers

1. Covariance is mean squared difference between each data point and the centre of the distribution measured by the mean
 2. Covariance is a data point that differs significantly from other observations
 3. Covariance is a statistical measure that expresses the extent to which two variables are linearly related
 4. Covariance is a statistical tool that is used to determine the relationship between the movements of two random variables.
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13. What is outlier in statistics?

Answers

1. outlier is mean squared difference between each data point and the centre of the distribution measured by the mean
 2. outlier is a data point that differs significantly from other observations.
 3. outlier is a statistical measure that expresses the extent to which two variables are linearly related
 4. outlier is a statistical tool that is used to determine the relationship between the movements of two random variables.
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14. What will be the probability of getting odd numbers if a dice is thrown?

Answers

1. $1/2$
 2. 2
 3. $4/2$
 4. $5/2$
-

15. What is the probability of getting a sum as 3 if a dice is thrown?

Answers

1. $2/18$
 2. $1/18$
 3. 4
 4. $1/36$
-

16. The probability of getting two tails when two coins are tossed is

Answers

1. $1/6$
 2. $1/2$
 3. $1/3$
 4. $1/4$
-

17. What is the probability of getting atleast one head if three unbiased coins are tossed?

Answers

1. 7/8
 2. 1/2
 3. 5/8
 4. 8/9
-

18. If two dice are thrown together, what is the probability of getting an even number on one dice and an odd number on the other dice?

Answers

1. 1/4
 2. 3/5
 3. 3/4
 4. 1/2
-

19. In a box, there are 8 orange, 7 white, and 6 blue balls. If a ball is picked up randomly, what is the probability that it is neither orange nor blue?

Answers

1. 1/3
 2. 1/21
 3. 2/21
 4. 5/21
-

20. The probability that it will rain tomorrow is 0.85. What is the probability that it will not rain tomorrow

Answers

1. 0.25
 2. 0.145
 3. 3/20
 4. 1/5
-

21. _____ divides the data into four equal parts

Answers

1. Median
 2. Quartiles
 3. Mean
 4. None of the above
-

22. Which of the following is a characteristic of a mean?

Answers

1. The sum of deviations from the mean is zero
 2. It minimises the sum of squared deviations
 3. It is affected by extreme scores
 4. All of the above
-

23. A hypothesis which defines the population distribution is called?

Answers

1. Null Hypothesis

2. Statistical Hypothesis
 3. Simple Hypothesis
 4. Composite Hypothesis
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24. If the null hypothesis is false then which of the following is accepted?

Answers

1. Null Hypothesis
 2. Positive Hypothesis
 3. Negative Hypothesis
 4. Alternative Hypothesis.
-

25. A chance variation in an observational process is

Answers

1. Dispersion/ Variability
 2. Measurement error
 3. Random error
 4. Instrument error
-

26. Which of the following is NOT true?

Answers

1. Pandas use to Easy handling of missing data (represented as NaN) in floating point as well as non-floating point data
 2. Pandas use to Size mutability: columns can be inserted and deleted from DataFrame and higher dimensional objects
 3. Pandas use to Intelligent label-based slicing, fancy indexing, and subsetting of large data sets
 4. Pandas is not a dependency of statsmodels.
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27. NumPy stands for ____

Answers

1. Number Python
 2. Numerical Python
 3. Numbers in Python
 4. None of the above
-

28. A variable that assumes any value within a range is called

Answers

1. Discrete variable
 2. Continuous variable
 3. Independent variable
 4. Dependent variable
-

29. The mean of a distribution is 14 and the standard deviation is 5. What is the value of the coefficient of variation?

Answers

1. 60.4%
 2. 48.3%
 3. 35.7%
 4. 27.8%
-

30. Sum of dots when two dice are rolled is

Answers

1. A discrete variable
 2. A continuous variable
 3. A constant
 4. A qualitative variable
-

31. A specific characteristic of a population is called

Answers

1. Statistic
 2. Statistics
 3. Parameter
 4. Sample
-

32. The number of accidents in a city during 2011 is

Answers

1. Discrete variable
 2. Continuous variable
 3. Qualitative variable
 4. Constant
-

33. If the mean of 6 numbers is 41 then the sum of these numbers is

Answers

1. 250
 2. 246
 3. 134
 4. 456
-

34. Find the mean of 8, 5, 7, 10, 15, 21

Answers

1. 11
 2. 12
 3. 13
 4. 14
-

35. To calculate the median, all the items of a series have to be arranged in a/an _____.

Answers

1. Descending order
 2. Ascending order
 3. Ascending or descending order
 4. None of the above
-

36. The algebraic sum of the deviations of a frequency distribution from its mean is always,

Answers

1. zero
 2. less than zero
 3. a non-zero number
 4. greater than zero
-

37. _____ in statistics, is defined as the value that has higher frequency in a given set of values. It is the value that appears the most number of times.

Answers

1. Range
 2. Mean
 3. Mode
 4. Median
-

38. How can you handle missing or corrupted data in a dataset?

Answers

1. Drop missing rows or columns
 2. Assign a unique category to missing values
 3. Replace missing values with mean/median/mode
 4. All of the above
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39. What is meant by overfitting of data?

Answers

1. An Overfitting machine learning model is not a suitable model and will be obvious as it will have poor performance on the training data.
 2. Overfitting is more likely with parametric and linear models that have not flexibility when learning a target function
 3. Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data
 4. Overfitting refers to a model that can neither model the training data nor generalize to new data.
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40. _____ is not a measure of central tendency

Answers

1. Range
 2. Mean
 3. Mode
 4. Median
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